

2 The European badger, *Meles meles*, pictured in Fig. 2.1, is a mammal found in England.

Badgers can contract bovine tuberculosis (bTB) caused by the pathogen, *Mycobacterium bovis*.

- Researchers estimated that 5% of the badger population in England are infected with bTB.
- bTB can be spread between badger populations and other populations of wild animals and farmed animals.
- Herds of dairy cows produce milk for human consumption. If a dairy herd becomes infected with bTB, the pathogen can be transmitted to human populations in milk.
- A variety of measures have been introduced in an attempt to reduce the bTB rate in badgers and so prevent infection of dairy cows.
- These measures include vaccination and the removal of badger populations (culling).



Fig. 2.1

- (a) Suggest how researchers estimated the percentage of the badger population that was infected with bTB.

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(b) It is possible that wild badgers infect dairy cows with bTB.

A group of scientists investigated the correlation between bTB infections in badgers and bTB infections in dairy cows.

- 30 study areas in England were identified.
- Each of the areas was approximately 100 km².
- 10 of the 30 study areas were randomly selected and given the codes **A–J**.
- The number of infected cows **and** the number of infected badgers were counted in each area.
- The number of infected badgers was determined by testing for the presence of *Mycobacterium bovis* in tissue samples taken from the badgers.
- The number of infected cows was determined by testing for a response (inflammation) to the introduction of antigenic material into the skin.

Data was collected as shown in Table 2.1.

Table 2.1

data set	areas where counts were made	year when counts were made
1	A, B, C, E, F, G, H	1999
2	D, I, J	2001

The scientists expressed their data on the numbers of infected cows and infected badgers as the population **density** of infected cows and infected badgers.

- (i) Explain what is meant by the term population density **and** suggest why the data was expressed in this way.

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(ii) State **three** variables that were **not** standardised in this investigation.

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(c) The scientists decided to analyse their data on the occurrence of bTB in cows and badgers statistically.

(i) State a null hypothesis for this investigation.

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- (ii) The scientists plotted a scatter graph of the population density of infected badgers and infected cows as shown in Fig. 2.2.

The scientists drew a line of best fit on the scatter graph.

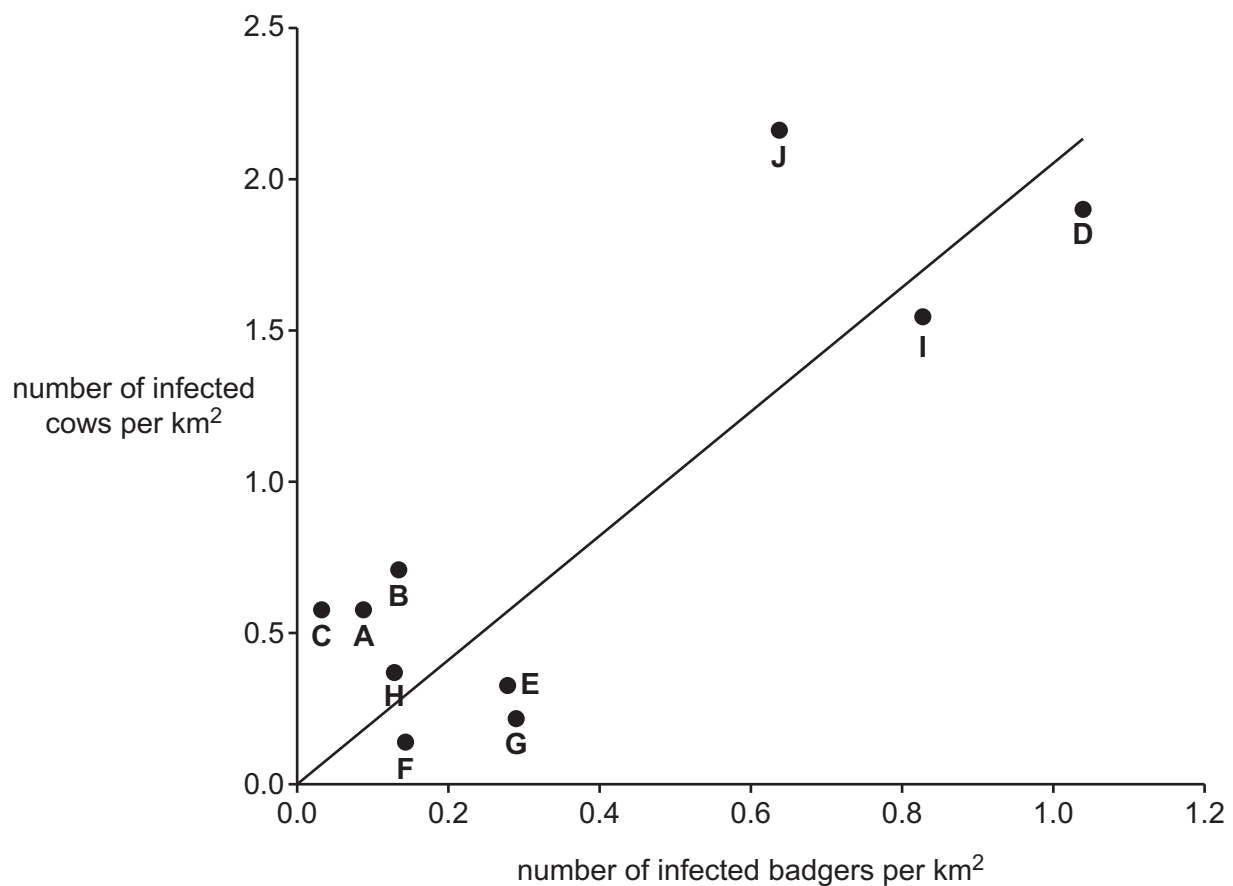


Fig. 2.2

The scientists carried out statistical tests on the two data sets combined and on each set separately.

The results of these tests are shown in Table 2.2.

Table 2.2

data set	sampling areas	correlation	significance at $p=0.05$
1 & 2	A, B, C, D, E, F, G, H, I, J	positive	significant
1	A, B, C, E, F, G, H	negative	not significant
2	D, I, J	negative	not significant

Some scientists suggested this information showed a strong positive relationship between the number of infected badgers in an area and the number of infected cows in the same area.

These scientists also suggested that bTB in herds of cows could be reduced by removing badgers through culling.

Evaluate whether or not the data in Fig. 2.2 and Table 2.2 support the suggestions these scientists made.

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